Worksheet 6.1 — Scatter Plots & Correlation

Grade 12 Mathematics of Data Management (MDM4U) · Unit 6: Two-Variable Statistics

The Big Ideas

A scatter plot of paired data shows *direction* (positive/negative), *form* (linear/non-linear), and *strength*. The correlation coefficient r measures *linear* strength, from -1 to 1.

- Explanatory (x) vs. response (y) variable.
- Describe a scatter by direction, form, and strength.
- $r \in [-1, 1]$: the sign gives direction, $|r|$ gives strength; r only measures *linear* association.

Concept in Focus

Two variables at once

A **scatter plot** graphs paired data as points to expose a relationship between two variables.

Reading correlation

An upward trend is **positive** correlation, downward is **negative**, and a shapeless cloud is none. A tight, line-like cloud is **strong**; a loose one is **weak**.

The correlation coefficient

The value r, from -1 to 1, measures the strength and direction of a *linear* relationship; values near ± 1 signal a strong linear fit.

Worked Examples

Example 1 — Explanatory vs. response

In the pair hours studied vs. test score, which is which?

Solution. Hours studied is explanatory (x); test score is the response (y).

Example 2 — Describe a scatter

Points climb steadily along a straight band. Describe the association.

Solution. Positive, linear, and strong.

Example 3 — Estimate r by eye

A tight, upward-sloping band of points. Estimate r.

Solution. Close to $+1$ (say $r \approx 0.9$).

Example 4 — Negative correlation

As outdoor temperature rises, heating cost falls. Describe r.

Solution. Negative — r is between -1 and 0.

Example 5 — No correlation

Shoe size vs. test score shows a shapeless cloud. Estimate r.

Solution. About 0 — no linear association.

Example 6 — Non-linear form

Points follow a clear U-shape. Is r a good summary?

Solution. No — r only measures *linear* strength and will understate a curved relationship.

Example 7 — Interpret a value

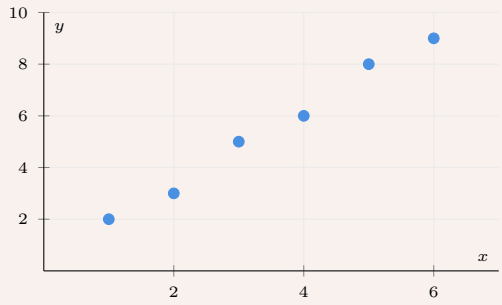
Interpret $r = -0.85$.

Solution. A strong *negative* linear relationship.

Example 8 — Read a scatter

Describe the plotted data.

Solution. A strong positive linear trend.



Example 9 — Compare r values (multi-part)

Given $r = -0.2,\ 0.95,\ -0.99,\ 0.5$, find
(a) the strongest;
(b) the weakest;
(c) the negative ones.

Solution. (a) -0.99 (largest $|r|$). (b) -0.2. (c) -0.2 and -0.99.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Which is the explanatory variable: hours of sunlight vs. plant growth?

Question 2

Describe the association when $r = 0.9$.

Question 3

Describe the association when $r = -0.4$.

Question 4

Which is stronger: $r = 0.8$ or $r = -0.9$?

Question 5

What does $r = 0$ indicate?

Question 6

A scatter curves sharply upward. Is r a good summary? Why?

Question 7

Estimate r for a tight, downward-sloping line of points.

Question 8

What is the likely sign of r for ice-cream sales vs. temperature?

Question 9

Rank by strength: $r = 0.3,\ -0.7,\ 0.95$.

Question 10

Which is explanatory: age vs. vocabulary size?

Question 11

Interpret $r = -0.99$.

Question 12

Two unrelated variables — what value of r do you expect?

Question 13 — Challenge

Given $r = 0.1,\ -0.6,\ 0.99,\ -0.95$, find
(a) the strongest;
(b) the weakest;
(c) the negative ones.

Question Recap & Answer Key

Question Recap

1. Which is the explanatory variable: hours of sunlight vs. plant growth?
2. Describe the association when $r = 0.9$.
3. Describe the association when $r = -0.4$.
4. Which is stronger: $r = 0.8$ or $r = -0.9$?
5. What does $r = 0$ indicate?
6. A scatter curves sharply upward. Is r a good summary? Why?
7. Estimate r for a tight, downward-sloping line of points.
8. What is the likely sign of r for ice-cream sales vs. temperature?
9. Rank by strength: $r = 0.3,\ -0.7,\ 0.95$.
10. Which is explanatory: age vs. vocabulary size?
11. Interpret $r = -0.99$.
12. Two unrelated variables — what value of r do you expect?
13. Given $r = 0.1,\ -0.6,\ 0.99,\ -0.95$, find
    (a) the strongest;
    (b) the weakest;
    (c) the negative ones.

✓ Answer Key

1. hours of sunlight
2. strong positive linear
3. weak-to-moderate negative
4. $r = -0.9$
5. no linear correlation
6. no — the relationship is non-linear
7. near -1
8. positive
9. $0.95 > 0.7 > 0.3$
10. age
11. very strong negative linear
12. ≈ 0
13. (a) 0.99; (b) 0.1; (c) $-0.6,\ -0.95$